

SP-III/Mathematics/301C-1C(T)/19

B.Sc. Semester III (General) Examination, 2018-19

MATHEMATICS

Course ID : 32118

Course Code : SPMTH-301C-1C(T)

Course Title : Algebra

Time: 2 Hours

Full Marks: 40

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

1. Answer *any five* questions: 2×5=10
- (a) Find the modulus and amplitude of $\frac{3+5i}{2-3i}$.
- (b) If a, b, c be positive real numbers, prove that
 $(a^2b + b^2c + c^2a)(ab^2 + bc^2 + ca^2) \geq 9a^2b^2c^2$.
- (c) If α be a multiple root of order 3 of the equation $x^4 + bx^2 + cx + d = 0$ ($d \neq 0$) show that
 $\alpha = -\frac{bd}{3c}$.
- (d) A relation R is defined on the set $\mathbb{Z}$ by " aRb if and only if $ab > 0$ " for $a, b \in \mathbb{Z}$. Examine if R is an equivalence relation.
- (e) Find the dimension of the subspace S of $\mathbb{R}^3$ where $S = \{(x, y, z): 2x + y - z = 0\}$.
- (f) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x^2 - 5$ and $g(x) = \frac{x}{x^2+1}$. Then find $f \circ g$ and $g \circ f$.
- (g) Find x such that the rank of $A = \begin{pmatrix} 2 & 1 & 4 \\ 1 & x & 2 \\ 4 & 0 & x+2 \end{pmatrix}$ is 2.
- (h) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$.
2. Answer *any four* questions: 5×4=20
- (a) If a, b, c, d be distinct positive real numbers and $S = a + b + c + d$ then prove that
 $\frac{S}{S-a} + \frac{S}{S-b} + \frac{S}{S-c} + \frac{S}{S-d} > 5\frac{1}{3}$
- (b) For what values of K , the following equations
 $x + y + z = 1, 2x + y + 4z = K, 4x + y + 10z = K^2$, have solutions and completely in each case.
- (c) Solve the equation $2x^4 - 5x^3 - 15x^2 + 10x + 8 = 0$, the roots being in geometric progression.

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SP-III/Math-301/C-1C/19

B.Sc. 3rd Semester (Programme) Examination, 2019-20

MATHEMATICS

Course ID : 32118

Course Code : SP/MTH-301/C-1C

Course Title: Algebra

Time: 2 Hours

Full Marks: 40

*The figures in the right hand side margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.**Unless otherwise mentioned, notations and symbols have their usual mainly.*

1. Answer *any five* questions: 2×5=10
- (a) Find the value of $(-1 + \sqrt{3}i)^{12}$.
- (b) Transform the equation to remove the square term from the equation $x^3 + 9x^2 + 15x - 25 = 0$
- (c) State first and second principle of induction.
- (d) Define rank of a matrix and hence find the rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.
- (e) If a, b, c be positive real numbers, not all equal, prove that $(a+b)(b+c)(c+a) > 8abc$.
- (f) A relation ρ is defined on the set $\mathbb{Z}$ by $a \rho b$ if and only if $a - b$ is divisible by 5 for $a, b \in \mathbb{Z}$. Show that ρ is an equivalence relation on $\mathbb{Z}$.
- (g) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 4x - 1$ and $g(x) = x^2 + 2$. Find $f \circ g$ and $g \circ f$.
- (h) Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x, y, z) = (x - y, 2z)$, show that T is linear.
2. Answer *any four* questions: 5×4=20
- (a) (i) Solve the equation $16x^4 - 64x^3 + 56x^2 + 16x - 15 = 0$. Whose roots are in arithmetic progression.
- (ii) Using Descartes' rule of sign, find the number of complex root of the equation $2x^4 + x^2 + 7x - 6 = 0$. 3+2=5
- (b) The matrix of a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to the ordered basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ is given by $\begin{pmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{pmatrix}$. Find T .
- (c) Find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$.
- (d) (i) Let a, b, c, d be positive real numbers, not all equal. Use Cauchy-Schwarz inequality to show that $(a+b+c+d)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) > 16$.
- (ii) If a, b, c be positive real numbers and $abc = k^3$, prove that
 $(1+a)(1+b)(1+c) \geq (1+k)^3$. 2+3=5

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(2)

- (d) (i) If a is prime to b , prove that $a + b$ is prime to ab .
- (ii) Prove that the product of any three consecutive integers is divisible by 6. 2+3=5
- (e) (i) Show that $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality.
- (ii) Let A, B be both finite sets of n elements and a mapping $f: A \rightarrow B$ is injective. Prove that f is a bijection. 3+2=5
- (f) Find the eigenvalues and the corresponding eigen vectors of the matrix
 $\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$.
3. Answer *any one* question: 10×1=10
- (a) (i) Solve: $x^3 - 12x + 65 = 0$
- (ii) Use the theory of Congruences to show that $7 \mid (2^{5n+3} + 5^{2n+3})$ for all positive integer n .
- (iii) Apply Descartes' rule of signs to find the nature of the roots of the equation $2x^4 + 14x^2 + 7x - 8 = 0$ 4+3+3=10
- (b) (i) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$ and find A^{-1} . 2+3=5
- (ii) State first principle of induction and using this principle prove that $2^n < n!$ for $n \in \mathbb{N}$ and $n \geq 4$. 2+3=5

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(2)

- (e) (i) Use first principle of induction to show that $7^{2n} + 16n - 1$ is divisible by 64, $\forall n \in \mathbb{N}$.
- (ii) Use theory of congruence, find the remainder when 3^{36} is divided by 77. 3+2=5
- (f) Solve the system of equations
 $x + 2y + z = 1$
 $3x + y + 2z = 3$
 $x + 7y + 2z = 1$
3. Answer *any one* question: 10×1=10
- (a) (i) Find the remainder when 3^{36} is divided by 77.
- (ii) Solve the equation using Cardan's method $x^3 - 27x - 54 = 0$.
- (iii) Solve the equation $x^8 + x^7 + x^6 + \dots + x + 1 = 0$ 2+5+3=10
- (b) (i) State Cayley-Hamilton theorem.
- (ii) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Hence find A^{-1} .
- (iii) Obtain a row-echelon matrix which is equivalent to $\begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix}$. 1+(3+2)+4=10

The figures in the margin indicate full marks

Notations and symbols have their usual meaning

1. Answer any five of the following questions:

(2x5=10)

- a) Using Principle of Mathematical Induction, prove that $3^{2n-1} + 2^{n+1}$ is divisible by 7 for all natural number n .
- b) Prove that $\frac{(n+1)^n}{2^n} > n!$
- c) Find the values of i^i
- d) Apply Descartes' rule of sign to examine the nature of roots of the equation $x^6 + 4x^4 + 2x^2 + 4x + 1 = 0$.
- e) State fundamental theorem of classical algebra.
- f) Find the dimension of the subspace W of R^3 defined by $W = \{(x, y, z) \in R^3 : x + y + z = 0\}$.
- g) Determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 3 & 0 \\ 6 & 2 & 3 \end{bmatrix}$$

- h) Use Cayley-Hamilton theorem to compute A^{-1} where

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

2. Answer any four of the following questions:

(5x4=20)

- a) If $\alpha = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ and if p is prime to n , prove that $1 + \alpha^p + \alpha^{2p} + \dots + \alpha^{(n-1)p} = 0$.
- b) If α be a root of the equation $x^3 - 3x - 1 = 0$, prove that the other roots are $2 - \alpha^2$ and $\alpha^2 - \alpha - 2$.
- c) How many different relations can be defined on a set with n elements? How many of these are reflexive? 3+2
- d) Determine the conditions for which the system of equations has (a) only one solution, (b) no solution, (c) many solution

$$x + y + z = 1$$

$$x + 2y - z = b$$

$$5x + 7y + az = b^2.$$

- e) Determine the linear mapping $T: R^3 \rightarrow R^2$ which maps the basis vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ of R^3 to the vectors $(1,1)$, $(2,3)$, $(3,2)$ of R^2 respectively. Hence find $T(6,1,1)$.
- f) Find the eigen values and the corresponding eigen vectors of the real matrix

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

3. Answer any one of the following questions:

(10 x 1 = 10)

- a) (i) Find the least positive residue in $2^{41} \pmod{23}$

(ii) If n be a positive integer, prove that $\frac{1}{2\sqrt{n+1}} < \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} < \frac{1}{\sqrt{2n+1}}$

(iii) Solve by Cardan's method $x^3 - 27x - 54 = 0$. 2+4+4

- b) (i) In R^2 , $\alpha = (3,1)$, $\beta = (2,-1)$. Determine $L[\alpha, \beta]$ and hence show that $L[\alpha, \beta] = R^2$.
- (ii) If α is an eigen value of a real orthogonal matrix A , then prove that $\frac{1}{\alpha}$ is also an eigen value of A .
- (iii) Obtain the fully reduced normal form of the matrix

$$\begin{pmatrix} 2 & 4 & 1 & 0 \\ 1 & 2 & 0 & 3 \\ 3 & 6 & 2 & 5 \end{pmatrix}$$

$$3+3+4=10$$

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22-23 / 22118

B.Sc. Semester-II Examination, 2022-23

MATHEMATICS [Programme]

Course ID : 22118

Course Code : SP/MTH/201/C-1B

Course Title : Algebra

[NEW SYLLABUS]

Time : 2 Hours

Full Marks : 40

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

Notations and symbols have their usual meaning.

UNIT-I

1. Answer any five from the following questions:

2x5=10

- a) Let a, b, c be three arbitrary elements of a group $(G, \cdot)$. If $a \cdot c = b \cdot c$, then show that $a = b$.
- b) Show that $3n(3n+1)^2 > 4((3n)!)^{\frac{1}{n}}$.
- c) State the Descartes' rule of signs.
- d) Simplify: $(1-i)\left(1-\frac{1}{i}\right)$
- e) Solve the equation $x^3 - 3x^2 + 4 = 0$ two of its roots being equal.

- f) If $a \mid c$ and $b \mid c$ with $\gcd(a, b) = 1$, then prove that $ab \mid c$

- g) Express $-1-i$ in polar form.

- h) Define order of an element in a group. In the group $(Z_6, +)$, find $o(\bar{1})$, $o(\bar{4})$ and $o(\bar{5})$.

UNIT-II

2. Answer any four from the following questions:

5x4=20

- a) i) If a, b, c are positive real numbers, then show that $a^3 + b^3 + c^3 \geq 3abc$
- ii) If a_1, a_2, a_3, a_4, a_5 be positive real numbers, then prove that

$$\left(\frac{a_1 + a_2 + a_3 + a_4 + a_5}{5} \right)^5 \geq \left(\frac{a_1 + a_2}{2} \right)^2 \left(\frac{a_3 + a_4 + a_5}{3} \right)^3$$

2+3

- b) i) Show that $\sinh(x + iy) = \sinh x \cos y + i \cosh x \sin y$.

- ii) Expand $\cos^7 \theta$ in a series of cosines of multiples of θ . 2+3

- c) i) Prove that the intersection of any two subgroups of a group $(G, *)$ is again a sub-group of $(G, *)$.

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(2)

- ii) Give an example with justification to show that the union of two sub-groups of a group need not be a sub-group of that group.

3+2

- d) i) Find the quotient and the remainder when $(3x^7 - x^6 + 31x^4 + 21x + 5)$ is divided by $(x+2)$.
- ii) Apply Descartes' rule of signs to find the nature of roots of the equation $x^4 + 16x^2 + 7x - 11 = 0$.
- e) i) If n be any positive integer, then prove that $n(n+1)(n+2)$ is divisible by 6.
- ii) Show that the square of an odd integer is of the form $(8k+1)$.
- f) Solve $x^3 - 18x - 35 = 0$ by Cardan's method.

UNIT-III

3. Answer any one of the following questions:

10×1=10

- a) i) State the fundamental theorem of algebra.
- ii) Find the roots of the equation $Z^8 = 1$
- iii) If x, y, z are positive real numbers and $x+y+z=1$, then prove that

$$8xyz \leq (1-x)(1-y)(1-z) \leq \frac{8}{27} \quad 1+4+5=10$$

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(3)

[Turn Over]

- b) i) Show that $(\mathbb{Z}_5, +)$ is a group.
- ii) Find the condition that the cubic equation $x^3 - px^2 + qx - r = 0$ should have its roots in G.P.
- iii) Find the remainder when $1!+2!+3!+\dots+100!$ is divisible by 16.

3+3+4=10

431/Math.

22-23 / 32118

B.Sc. Semester-III Examination, 2022-23

MATHEMATICS [Programme]

Course ID : 32118

Course Code : SP/MTH/301/C-1C

Course Title : Algebra

Time : 2 Hours

Full Marks : 40

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

Notations and Symbols have their usual meanings.

UNIT-I

1. Answer any five of the following questions:

2×5=10

- a) Solve the equation $x^4 + x^2 - 2x + 6 = 0$, where $1+i$ is one root of the given equation.
- b) If n is a positive integer, prove that
$$\frac{1.3.7\dots(2^n-1)}{2.4.8\dots 2^n} < \frac{2^n}{2^{n+1}-1}$$
- c) Find the product of all the values of $(1+i)^{\frac{4}{5}}$.
- d) Give an example of an infinite set S and a mapping $F:S \rightarrow S$ such that F is surjective but not injective.

[Turn Over]

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- e) Show that the relation ρ defined on $\mathbb{R}$ by the rule " $x\rho y$ iff $x-y$ is irrational" is not an equivalence relation.
- f) Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be two invertible functions. Then show that $g \circ f$ is invertible.
- g) Determine the rank of the matrix $A^3 + A^2 + A$, where

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 3 & 0 \\ 6 & 2 & 3 \end{bmatrix}$$

- h) Prove that $n^2 - 2$ is not divisible by 4, $n \in \mathbb{Z}$.

UNIT-II

2. Answer any four of the following questions:

5×4=20

- a) Use De Moivre's theorem to prove that

$$\tan 5\theta = \frac{5\tan\theta - 10\tan^3\theta + \tan^5\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$$

- b) Solve the equation : $x^4 + x^3 - 2x^2 + 4x - 24 = 0$.
- c) Suppose A is a 2×2 real matrix with trace 5 and determinant 6. Find the eigenvalues of the matrix $B = A^2 - 2A + I_2$.

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(2)

- d) Find a linear operator T on $\mathbb{R}^3$ such that $\text{Ker } T$ is the subspace $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ of $\mathbb{R}^3$.
- e) i) Prove that the rank of a real skew-symmetric matrix cannot be 1.
- ii) If $\gcd(m, n) = 1$, prove that $\gcd(mn, m+n) = 1$, where m and n are positive integers. 3+2
- f) For what values of k the following system of equations has a non-trivial solution? Solve in any one case.
- $$x + 2y + 3z = kx; 2x + y + 3z = ky; 2x + 3y + z = kz$$

UNIT-III

3. Answer any one of the following questions:

$$10 \times 1 = 10$$

- a) i) Prove that two eigenvectors of a square matrix A over the field $\mathbb{R}$ corresponding to two distinct eigen values of A are linearly independent.
- ii) Find the sum of 99^{th} powers of the roots of the equation $x^7 - 1 = 0$.

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[Turn Over]

B.Sc.-II/Mathematics-22118

B. Sc. Semester II Examinations, 2023-24

Subject: Mathematics

Course ID: 22118

Course Code: SP/MTH/201/C1-B

Course Title: Algebra (NEW CBCS Syllabus)

Time: 2 Hours.

Full Marks: 40

The figures in the right-hand margin indicate marks.

Candidates are required to give their answer in their own words as far as practicable.

Notations and Symbols have their usual meaning.

1. Answer any 5 (five) of the following questions:

$$(2 \times 5 = 10)$$

- a) Prove that the sum of 99^{th} powers of the roots of the equation $x^7 = 1$ is 0.
- b) Find the value of $\phi(2401)$.
- c) If $S_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then show that $(1+n)S_n > 2n$.
- d) A relation R is defined on $\mathbb{Z}$ by aRb iff $ab < 0 \forall a, b \in \mathbb{Z}$. Verify whether R is an equivalence relation.
- e) If n is odd then determine the exact number of real roots of the equation $x^n - 1 = 0$.
- f) Define linear transformation from one space to another. When is it called zero transformation?

[Turn Over]

- iii) Find the least positive residue of $3^{36} \pmod{77}$. 4+3+3=10

- b) i) Show that $S = \{(2, -5, 3)\}$ is not a subspace of $V_3(\mathbb{R})$ generated by the vectors $(1, -3, 2)$, $(2, -4, 1)$, $(1, -5, 7)$.

- ii) Show that the matrix $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ is not

diagonalizable.

- iii) Solve the equation:

$$x^6 - x^5 + x^4 - 2x^3 + x^2 - x + 1 = 0.$$

$$3+3+4=10$$

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- g) What is the fundamental theorem of arithmetic?

- h) Prove that a set of vectors containing the zero vector can't be linearly independent.

UNIT II

2. Answer any 4 (four) of the following questions:

$$(5 \times 4 = 20)$$

- a) State Cayley-Hamilton's theorem. Find the eigen values of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. 2+3
- b) For what value of k the system of linear equations $x + y + z = 2$, $2x + y + 3z = 1$, $x + 3y + 2z = 5$ is solvable and then solve it. $3x - 2y + z = k$
- c) Find a basis for $\mathbb{R}^3$ that contains the vectors $(2, 4, 0)$ and $(2, 6, 2)$. 2+3
- d) Two mappings $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2$ and $g(x) = x - 2 \forall x \in \mathbb{R}$. Then verify that $g \circ f \neq f \circ g$ with usual notations. 5
- e) Prove that $\sqrt[n]{i} + \sqrt[n]{-i} = 2 \cos \frac{\pi}{2n}$ where n is a positive integer. 5
- f) Solve $x^3 - 3x + 2 = 0$ by Cardan's method. 5

UNIT III

3. Answer any 1 (one) of the following questions:

(10 × 1 = 10)

a) (i) If $\alpha + i\beta$ ($\beta \neq 0$) is a root of the equation $x^3 + qx + r = 0$ then find the other roots of it.

(ii) What do you mean by $a \equiv b \pmod{m}$? If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then prove that $a \equiv c \pmod{m}$.

(iii) Solve the bi-quadratic

$x^4 + 2x^3 - 21x^2 - 22x + 40 = 0$ given that the sum of two roots of it is equal to the sum of the other two roots.

2 + 2 + 6

b) (i) If V & W are two sub-spaces of R^n and $T: V \rightarrow W$ is a linear transformation then prove that $T(\theta_V) = \theta_W$.

(ii) Use method of induction to establish $2n + 1 \leq 2^n \forall n \geq 3, n \in \mathbb{N}$.

(iii) Reduce the matrix $A = \begin{pmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 6 \end{pmatrix}$ to normal form and hence find $\text{Rank}(A)$.

3 + 2 + 5

f) Use Cayley-Hamilton theorem to compute A^{-1} where,

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

g) Determine the rank of the matrix A^2 where

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 3 & 0 \\ 6 & 2 & 3 \end{bmatrix}$$

h) Find the dimension of the subspace W of R^3 defined by

$$W = \{(x, y, z) \in R^3 : x + y + z = 0\}$$

UNIT-II

2. Answer any four from the following questions.

4 × 5 = 20

a) If one root of the equation $x^3 + ax + b = 0$ is twice the difference of the other two, then

prove that one root is $\frac{13b}{3a}$.

b) i) Show that $11.3! .5! \dots (2n-1)! > (n!)^n$.

ii) Prove that number of primes is infinite.

3 + 2

MATHEMATICS

Course ID : 32118

Course Code : SP-MTH-301/C-1C

Course Title : Algebra

[Syllabus - 2017]

Time : 2 Hours

Full Marks : 40

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer any five from the following questions:

5 × 2 = 10

a) State fundamental theorem of classical algebra.

b) Prove that $\frac{(n+1)^n}{2^n} > n!$

c) Find the values of i^i .

d) Show that the function $f: R \rightarrow R$ defined by $f(x) = |x| + x$ is neither injective nor surjective.

e) Show that the relation p defined on R by the rule " xpy iff $x + y$ is irrational" is not an equivalence relation.

c) How many different relations can be defined on a set with n elements? How many of these are reflexive? Give reason.

3 + 2

d) Find a linear operator T on R^3 such that $\text{Ker } T$ is the subspace

$$U = \{(x, y, z) \in R^3 : 2x + y - z = 0\} \text{ of } R^3.$$

e) Find two integers u and v satisfying $63u + 55v = 1$.

3 + 2

f) If α be an eigen value of a real orthogonal matrix A , then prove that $\frac{1}{\alpha}$ is also an eigen value of A .

UNIT-III

3. Answer any one from the following questions:

1 × 10 = 10

a) i) Prove that the eigen values of a real skew-symmetric matrix are purely imaginary or zero.

ii) Find the least positive residue in $2^{41} \pmod{23}$.

5 + 5 = 10

b) i) If $(1+x)^n = a_0 + a_1x + a_2x^2 + \dots$, prove that

$$a_0 + a_4 + a_8 + \dots = 2^{n-2} + 2^{\frac{n-2}{2}} \cos \frac{n\pi}{4}$$

ii) A mapping $T : R^3 \rightarrow R^3$ is defined by

$$T(x, y, z) = (x + y + z, 2x + y + 2z, x + 2y + z), (x, y, z) \in R^3$$

Show that T is a linear mapping. Find Ker

T and the dimension of Ker T.

$$5+5=10$$